

External Residual Adversarial Re-audit --- Paper I, preprint_draft_v1.3

Claude Opus 5 (claude-opus-5), Anthropic

2026-08-21

External Residual Adversarial Re-audit — Paper I, preprint_draft_v1.3

Specification: RESIDUAL_AUDIT_REQUEST_SPEC.md Specification SHA-256:
e8d92cf5fffb55b02eada99e4e446207752dbe28f1b8492195838b123d85435 Run: run_2026-08-21_v1.3
Audit class: external residual adversarial re-audit Audit date: 2026-08-21

1. Executive verdict

PASS_WITH_RESIDUALS

What this means. All nine mandatory correction controls pass, the general regression passes in full, and there is **no unresolved BLOCKER** and **no unresolved MAJOR**. The MAJOR finding that made v1.2 fail — five duplicated Spanish blocks reaching the delivered PDF — is **fully corrected and independently re-verified**.

Control 6 was initially assessed **PARTIAL**, because the newly cited Chen–Erdős–Ordman author-hosted scan could not be retrieved by any of four routes the auditor attempted. The owner then supplied the scan, downloaded from the cited URL in their own browser. The auditor rendered and read it, and the 3/16 constant is now **verified against the primary source with a pinpoint**: Theorem 1, page 23. Control 6 therefore passes. This evidence is disclosed as **owner-supplied and auditor-verified**, not as an independent auditor retrieval (Section 4.4).

Why not a plain PASS. Two correctable MINOR package defects remain open: compiler residue left inside the sealed package, and two transposed declaration namespaces in the changelog (Section 7). Neither is claim-critical, and Specification Section 7 permits minor findings that are compatible with the PASS rule — but a sealed package carrying stray build artifacts is a real defect the owner can fix in minutes. **Deleting tmp/ and correcting the two changelog names would make a plain PASS available on a new target hash.** Standing external residuals also remain (Section 10).

This audit is not human peer review and does not prove global novelty.

2. Target identity — all eight anchors verified

Recomputed from the delivered bytes before any analysis. **8/8 match.**

Artifact	Required SHA-256	Result
English Markdown	f3094b670c93ff622c3f573cdab61bdd0f5d84007f04b1888b364e0183565bea	MATCH
Spanish Markdown	57ba967fc2de805b7bbb4cf5f937727bde43c2ada80fa794bcb5a727db05b8b	MATCH
English LaTeX	1a87de70548879ca90a714ec9e1b10c8576b380a749785916e5d59176e479465	MATCH
Spanish LaTeX	f83ee709c1168b5b9afe504e6b6a43763bf3bb2a08f674f2031f83670ba9bb56	MATCH
English PDF	7d04c47692b613c8d6e2cc4471f0205128b4ae06ca11d581a08d020eb2236db0	MATCH
Spanish PDF	3ad16ae86e8fcad358bf964a6ae98ae053b0bde2fdf3140e008cedab78b90c2a	MATCH
Lean archive	0181506408644fc1f8872d711de5a98a500f4052aa295bcd6f8c82776694fd3a	MATCH
Internal-audit ZIP	f261add8050f1ab8779d9e25f345475790768368f4c056067358153db6639382	MATCH

Input freeze sealed before analysis: **221 files, 24,675,148 bytes**, aggregate
a8852b0e4d424615d299d9b1dba5bb0aef244295193a662f57afc48a652427fb, with the audit-output sub-

tree excluded. Manuscript sidecar is **LF-only, 6/6 entries verify**. No path resolves into **superseded/**. The target did not change during the run and was not modified by the auditor.

3. What actually changed from v1.2

Established by a line-level diff of the two English manuscripts, not taken on trust: **14 changed regions, every one of them an announced correction**. v1.2 had 1,089 lines; v1.3 has 1,101.

No theorem statement, no constant, no proof step changed, with two exceptions that both *strengthen* the argument: the added H -slack accounting in the derivation of (4.7), and the domain restriction on the Appendix A.2 excess formula. This matches the specification's own statement in Section 1 and the `CHANGELOG_v1.3.md`.

4. Mandatory correction controls, one by one

#	Finding	Verdict	Evidence
1	EXT-P1-L-001 duplicated Spanish blocks	PASS	Section 5
2	EXT-P1-M-001 integrity metadata	PASS	Section 6
3	EXT-P1-D-001 derivation of (4.7)	PASS	Section 4.1
4	EXT-P1-E-002 Appendix A.2 restriction	PASS	Section 4.2
5	EXT-P1-J-001 meaning of "sharp"	PASS	Section 4.3
6	EXT-P1-I-001 CEO scan and 3/16	PASS	Section 4.4
7	EXT-P1-I-002 Schrijver pinpoint absent	PASS	Section 4.5
8	EXT-P1-I-003 stale repository citation absent	PASS	Section 4.5
9	P1-IA-V13-001 no residual pinpoint	PASS	Section 4.5

4.1 Control 3 — the derivation of (4.7) is now complete and correct

Both halves of the v1.2 finding are addressed.

- **The substitution is stated.** English and Spanish L270 now read that the second substitution "gives $z_e = x_e$ and leads to the fixed **capped-dual** polytope". In v1.2 the composition $y = 1-x$ then $z = 1-y$ was left unremarked, so (4.5) appeared to be a new polytope rather than the original capped dual.
- **Every edge of H is accounted for.** New text at L303–317 supplies the cancellation v1.2 omitted:

$$|w_1| = b_{\geq 2}/2 + \sum_{e \in H} (1 - \kappa_e) \quad |w_2| = M(\kappa) - \sum_{e \in H} (1 - \kappa_e)$$

and notes explicitly that "On H , the residual coefficient r_e is zero" — precisely the distinction the v1.2 finding turned on, since $r_e = \max\{1-\kappa_e, 0\}$ vanishes on H rather than being negative.

Independently verified symbolically, not merely read:

- the new $|w_1|$ identity is *consistent with* (4.3) $|w_1| = |H| + \sum_{e \in L} \kappa_e$, given Lemma 3.1 (sum over all e of $\kappa_e = b_{\geq 2}/2$). Exact identity, residual zero.
- adding the two new identities yields (4.7) $V_{\text{com}} = b_{\geq 2}/2 + M(\kappa)$ exactly: the two $\sum_H (1 - \kappa_e)$ terms cancel. Exact identity, residual zero.

So the accounting is not just present, it is right.

4.2 Control 4 — Appendix A.2 is now correctly restricted

L888 now reads "If $A \geq 0$ and $o \geq 3$ ", and L904–906 route the cases $o = 0, 1, 2$ to Appendix A.3. Verified symbolically that this is the *exact* correct domain:

Endpoint	Excess over U	Matches printed $(B-2A)/6$?
$(0, 1/2, 1/3)$, valid when $o \geq 3$	$(B - 2A)/6$	yes
$(0, 1/2, 0)$, the case $o \leq 2$	$(B - 2A - 2C)/6$	no , differs by $-2C/6$

The v1.2 text printed $(B-2A)/6$ unconditionally, which is false for $o \leq 2$. The v1.3 restriction is precisely the right one.

4.3 Control 5 — "sharp" no longer overreaches

The phrase "the sharp target" is **gone** from L691/L705. "Sharp" now appears only inside the declaration names `paperI_main_sharp` and `assembly_sharp`, and L750 still states plainly that "The additive term $n/2$ in Theorem 1.1 is not optimized", giving the $n/6$ to $n/2$ window. The **CHANGELOG** records the intent as clarifying that **sharp** refers to the quadratic coefficient $1/6$. Consistent, and consistent with this audit's own computation that the extremal family attains $n^2/6 + n/6$, which is what makes $n/2$ non-optimal.

4.4 Control 6 — cited, and now verified against the primary source: PASS

The citation. Reference [2] now reads, in both languages and in all four generated artifacts: "...World Scientific, 1994, pp. 21-30, author-hosted scan: https://ordman.net/MathResearch/CEOClique_Parts.pdf, accessed August 21, 2026." The URL is present once in the English Markdown, Spanish Markdown, both LaTeX sources **and both rendered PDFs** - the acceptance criterion the owner's own correction matrix set.

Retrieval, and how the evidence was obtained. The auditor could not retrieve the scan. Four routes were attempted and all failed:

Route	Outcome
WebFetch on the printed <code>https</code> URL	certificate expired
<code>curl</code> on the <code>http</code> URL	HTTP 302 to <code>https</code> , certificate expired
internal browser, <code>https</code>	navigation denied
<code>web.archive.org</code> snapshot	not reachable from this environment

The auditor declined to disable certificate validation. **The owner then supplied the file**, downloaded from the cited URL in their own browser after clicking through the certificate warning. This is disclosed plainly: the evidence below is **owner-supplied and auditor-verified**, not an independent auditor retrieval. The auditor did perform the verification itself - the file is an image-only scan (`pdftotext` extracts zero lines, Producer "EPSON Scan"), so it was rendered at 150 dpi and read.

Received file SHA-256 `b7d965e6e00cd664ad2249ad7fa3640330009c3c9057301d73f752a718b28262`, 6 pages, archived in `20_EVIDENCE/I_CITATIONS_NOVELTY/results/`.

Identity confirmed. Title "CLIQUE PARTITIONS OF SPLIT GRAPHS"; authors Guan-Tao Chen (North Dakota State University), Paul Erdos (Mathematical Institute, Hungarian Academy Of Sciences), Edward T. Ordman (Memphis State University). A handwritten annotation on the first page records the volume: "Combinatorics, Graph Theory, Algorithms, and Applications, ed. Yousef Alavi, Don Lick, Jiuqiang Liu; World Scientific Pub Co. Pte Ltd, Singapore, 1994 (Conference Beijing 1-5 June 1993)". Printed folios run -21-, -22-, -23-, -24-, consistent with the cited "pp. 21-30".

The 3/16 constant, verified twice.

- Abstract, page 21, verbatim: "It is well-known that edge-partitioning a split graph on n vertices may require as many as $n^2/6 + n/6$ cliques. We show that $(3/16)n^2 + 0(n)$ cliques will always suffice."
- **Pinpoint**, page 23, Section 2 "Statement of results": "**Theorem 1. For all split graphs G_n , $cp(G_n) \leq (3/16) n^2 + 0(n)$.**" with "Corollary 1. The same bound applies to threshold graphs."

The manuscript's attribution is therefore **SUPPORTED, with pinpoint Theorem 1, page 23**. The residual that made this a MINOR finding in the v1.2 report (EXT-P1-I-001) is **closed**.

Two further findings of independent value, from the same source.

1. Page 22, Example 1: "The clique partition number of $G_n = K_n - K_{2n/3}(\text{bar})$ is $n^2/6 + n/6$, provided 6 divides n ." This is **exactly** the complete-split benchmark value this audit verified independently by explicit construction for $p = 2..40$ in the Paper III Gate F evidence. Primary source and independent computation agree.
2. Page 23: "A split graph on n vertices may need $n^2/6 + 0(n)$ cliques to partition it. **We cannot show that this many will suffice.**" The original authors thus flag as open precisely the gap Paper III claims to close. Their Theorem 2 obtains $n^2/6 + 0(n)$ only for the special form $K_n - K_m(\text{bar})$, a difference of cliques, **not** for general split graphs. Paper III's general claim is therefore genuinely stronger than CEO Theorem 1 and is not implied by CEO Theorem 2 - no collision, and the $3/16 \rightarrow 1/6$ framing is well posed.

What remains. The printed URL is still unusable as printed, since the host serves an expired certificate. That is now an accessibility matter rather than an evidentiary gap, recorded as RES-V13-004 (NOTE).

4.5 Controls 7, 8, 9 — removals verified across all six artifacts

Check	EN md	ES md	EN tex	ES tex	EN pdf	ES pdf
Corollary 7.1g pinpoint	absent	absent	absent	absent	absent	absent
github.com/jtraverso citation	absent	absent	absent	absent	absent	absent

Schrijver is now cited at the **verified chapter level**: "[4, Chapter 7]" at L1025, and "Schrijver [4, Chapter 7] remains the classical reference" at L781. Control 9 is therefore satisfied: no residual occurrence of the old pinpoint survived regeneration.

The bibliography went from 10 entries to 9 with the removal of the repository entry. **No dangling and no orphaned citation resulted**: keys cited in the body are exactly $\{1,2,3,5,6,7\}$ plus [4] cited as "[4, Chapter 7]", and the reference list defines $\{1,\dots,7\}$. Two apparent anomalies were investigated and dismissed — an apparent citation "[0]" is the unit cube $\mathbf{z}$ in $[0,1]^{\mathbf{E(K)}}$ in (4.5), and [4] appeared "uncited" only because the pinpoint form escapes a plain-bracket regex.

5. Control 1 — duplication, tested generally rather than by known phrase

The specification requires an exact **and near-duplicate** scan of normalized long units in all six artifacts, manual inspection of every candidate, and a page-image check. All were performed.

Method. Units are (a) paragraphs, i.e. maximal runs of non-blank lines, and (b) individual lines. Normalization: NFKC, lowercase, whitespace collapsed, punctuation stripped. Threshold: normalized length ≥ 60 characters. Near-duplicate criterion: `diff`lib ratio ≥ 0.92 between distinct units. **Exclusions: none** — every long unit in every artifact was compared against every other. Pages: exact SHA-256 of the rendered PNG plus a 64x88 grayscale signature compared by mean absolute difference.

Result.

Artifact	Long units (para/line)	Exact duplicate classes	Near-duplicate pairs
EN Markdown	116 / 106	0 / 0	1 / 1
ES Markdown	121 / 111	0 / 0	1 / 1
EN LaTeX	133 / 133	0 / 0	1 / 3
ES LaTeX	139 / 139	0 / 0	1 / 3
EN PDF text	68 / 243	1 / 0	0 / 0
ES PDF text	73 / 274	1 / 0	0 / 0

Every candidate was inspected individually. All are legitimate:

- the near-duplicate paragraph in each artifact is the constraint set (6.4) $3a \geq 1$, $a+2b \geq 1$, $2b+g \geq 1$, $3g \geq 1$ restated in Appendix A.1 — the two differ only by the `\tag{6.4}` marker;
- the near-duplicate lines are the two *different* declarations `paperI_main_sharp` and `paperI_main` reporting the same axiom footprint, and in the LaTeX the pair `\pdftitle{...}` / `\title{...}`, both of which are required;
- the one "exact duplicate" in each PDF text layer is the string `propept`, `Classical.choice`, `Quot.sound` on consecutive lines — adjacent table cells for different declarations that legitimately share a footprint, flattened into separate lines by `pdftotext`.

Page images.

PDF	Pages	Exact-duplicate page sets	Min signature distance	Median	Pairs below strict threshold 1.5
English	19	0	3.850	8.496	0
Spanish	20	0	4.619	8.515	0

A genuinely repeated page would show a signature distance near zero; the closest pair in either document sits at roughly half the median separation, nowhere near identity. No near-blank page and no anomalous black area in either document.

The five v1.2 duplications are gone and nothing replaced them. For the record, they were at Spanish v1.2 lines 51–65/67–81, 730/732, 742/744, 758–762/764–768 and 776–785/787–794, including an orphaned sentence and an ungrammatical duplicate "Esto prueba Teorema 1.1."; none of these patterns occurs in v1.3.

Note: the Spanish PDF remains 20 pages against the English 19, unchanged from v1.2. Since the duplicated content has been removed and the page count did not fall, the page delta is ordinary Spanish text expansion plus the newly added H-accounting and A.2 text — not duplication.

6. Control 2 — integrity metadata now identifies v1.3 and verifies exactly

28 of 28 sidecars in the package verify by content, with zero problems. This is the control v1.2 failed.

Sidecar	Entries	Result
01_manuscript/..._v1.3_SHA256.txt	6	all verify, LF-only
04_integrity/CURRENT_TARGET_SHA256.txt	8	all verify
04_integrity/INITIAL_SOURCE_SHA256.txt	2	both verify (in v1.2, 1 of 2 was unresolvable)
05_formalization/.../PACKAGE_MANIFEST.sha256	31	all verify
05_formalization/.../SOURCE_MANIFEST.sha256	19	all verify
internal-audit manifests (v1.2 baseline + current)	40 across 22 files	all verify

04_integrity/ now carries v1.3-specific records: CURRENT_TARGET_SHA256.txt, EXTERNAL_AUDIT_CORRECTION_MATRIX.md, SEMANTIC_INTEGRITY_REPORT_v1.3.md. The stale "Pending for v1.1" baseline that produced EXT-P1-M-001 is gone.

7. New findings opened against v1.3

ID	Severity	Status	Summary
RES-V13-001	MINOR	OPEN	stray tmp/internal_report_v1.3/ directory containing compiler residue
RES-V13-002	MINOR	OPEN	CHANGELOG_v1.3.md names two declarations with swapped namespaces
RES-V13-003	MINOR	RESOLVED	the 3/16 constant is verified against the primary source (Theorem 1, page 23)
RES-V13-004	NOTE	OPEN	the cited URL is unusable as printed; the host serves an expired certificate

RES-V13-001 — compiler residue inside the frozen package

The package contains tmp/internal_report_v1.3/ with three files: INTERNAL_AUDIT_FINAL_REPORT.aux (628 B), .log (17,594 B) and .pdf (164,176 B). The .log and .pdf are **byte-identical** to files already correctly placed under 02_validation/01_INTERNAL_AUDITS/10_REPORT/, so tmp/ is pure LaTeX scratch left behind by the internal-report build. Specification Section 6 explicitly requires verifying "no compiler residue"; this is that. **Disposition:** delete the tmp/ subtree and reseal.

RES-V13-002 — swapped namespaces in the changelog

CHANGELOG_v1.3.md line 10 states the axiom query was expanded to cover PaperI.Split.assembly_sharp and PaperI.residual_duality. The actual declarations are PaperI.assembly_sharp and PaperI.Split.residual_duality — the namespaces are transposed. The **manuscript itself is correct**: Appendix C prints the four declarations with the right namespaces, matching this auditor's verbatim #print axioms output exactly. So the error is confined to package documentation and does not affect any claim. **Disposition:** correct the two names in the changelog.

RES-V13-003 — **RESOLVED**

The 3/16 constant is verified against the primary source with a pinpoint (Theorem 1, page 23). See Section 4.4. No further action.

RES-V13-004 — the printed URL does not resolve (**NOTE**)

ordman.net redirects http to https and serves an expired certificate, so a reader following reference [2] hits a certificate error. Now an accessibility matter, not an evidentiary gap. **Disposition:** optionally note the expired certificate in the manuscript, or cite a stable mirror alongside the author-hosted copy. The defect is on a third-party host and cannot be fixed by editing the target.

8. General regression (specification Section 6)

8.1 Mathematics — 14 of 14 symbolic identities exact; 6,732 LP cases with zero mismatches

Every identity is checked as an exact polynomial identity (expand, subtract, require the residual to be identically zero), not sampled. Two of the fourteen are new in v1.3 and are the ones that validate the control-3 correction.

Tag	Identity	Holds
v13-4.3/4.8	v1.3 ‘	w_1

Tag	Identity	Holds
v13-4.7	the two v1.3 identities add to (4.7); <code>sum_H</code> terms cancel	yes
L3.1	$C(d,2)/(d-1) = d/2$	yes
6.3	$A+B+C = (p^2-p-sq)/2$	yes
7.2, 7.4, 7.4c, 7.6	the three Section 7 comparisons and the nonnegativity certificate	yes
8.2, 8.3, 8.res	the simplification, the assembly difference, the residual regime	yes
9.cs	complete-split value $n^2/6 + n/6$ at $n = 3p$	yes
A.2-o>=3, A.2-o<=2	the A.2 excess at each endpoint	yes

Orbit program, exact rational vertex enumeration, $p = 2..18$, $q = 1..44$, $s = 2..p$: **6,732 cases, 0 closed-form mismatches, 0 violations of (7.8).**

Tightness Remark, all claims re-verified with the same sharp constants:

Regime	Minimum slack	At	Claim
$s \geq 3, o \geq 3$	9/4	(6,3,3)	$\geq 9/4$ — PASS, tight
$s = 2, o \geq 1$	1/4	(3,3,2)	$\geq 1/4$ — PASS, tight
$o = 1$	1/4	(3,3,2)	$\geq 1/4$ — PASS, tight
$o = 2$	1	(4,2,2)	$\geq 1/4$ — PASS
$o = 0$	0	(2,4,2)	equality regime

The mandated point $(p,q,s) = (2,4,2)$ gives LP value -3 and slack **exactly 0: CONFIRMED** in the equality family. All 17 equality cases have $o=0$ and $q=2p$, exactly as the Remark claims.

A direct enumeration of split graphs with `nu_3*` by exact rational simplex was launched to re-falsify Theorem 1.1 itself; its result file is recorded under `20_EVIDENCE/MATH_REGRESSION/results/`.

8.2 Bilingual chain

All eight corrections are present in **both** languages, at **identical line numbers** (270, 308/311/314, 888, 904/906, 1091/1089, 781/1025), which indicates a properly regenerated pair rather than a hand-patched one. Zero self-containment violations: neither manuscript references v1.1.4, v1.1.5, "supersede", the correction matrix, the external or internal audit reports, the changelog, or any EXT-P1-* finding id. The manuscript is self-contained, as Section 6 requires.

8.3 Rendered artifacts

Both PDFs: LuaTeX-1.24.0, A4, unencrypted, text layer Unicode-recoverable including mathematical alphanumeric symbols. All 39 pages (19 EN + 20 ES) rendered at 100 dpi and analysed; no blank page, no anomalous black area, no duplicated page, no malformed structure.

8.4 Package hygiene

Check	Result
zero-byte files	0
stray <code>\$o</code>	absent
paths into superseded/	0
compiler residue	3 files — RES-V13-001
stale version, path or hash in the manuscripts	none
truncated report or script	none found

9. Gate H — reused under the byte-identity rule

Labelled `REUSED_BYTE_IDENTICAL_EXTERNAL_EVIDENCE`. All five conditions of specification Section 5 are shown:

1. **Archive hash.** The delivered archive is 0181506408644fc1f8872d711de5a98a500f4052aa295bcd6f8c82776694fd3a, exactly the required value.
2. **Byte-identical to what was previously rebuilt externally.** The archive was re-extracted and compared member by member against the external v1.2 clean room: **32 of 32 members byte-identical, 0 differing, 0 missing.**
3. **Nothing that governs the build changed.** `lean-toolchain`, `lake-manifest.json`, `lakefile.toml` and `FreezeAxioms.lean` are each **IDENTICAL** to the versions built. Toolchain Lean 4.28.0 commit 7e01a1bf5c70fc6167d49c345d3bf80596e9a79b; Mathlib pinned at 8f9d9cff6bd728b17a24e163c9402775d9e6a365.

4. **The v1.3 manuscript's formal claims check out against those unchanged sources.** Appendix C's axiom report lists four declarations; all four match the auditor's own independently obtained verbatim output, **including the namespaces** that the changelog gets wrong:

```
PaperI.Split.paperI_main_sharp depends on axioms: [propext, Classical.choice, Quot.sound]
PaperI.paperI_main             depends on axioms: [propext, Classical.choice, Quot.sound]
PaperI.assembly_sharp          depends on axioms: [propext, Classical.choice, Quot.sound]
PaperI.Split.residual_duality depends on axioms: [propext, Classical.choice, Quot.sound]
```

Appendix C's stated theorem $G.\text{Phi} \leq (G.n : \mathbb{R})^2/6 + (G.n : \mathbb{R})/2$ matches the compiled `PaperI.Split.paperI_main_sharp` : forall (G : PaperI.Split), $G.\text{Phi} \leq G.n^2/6 + G.n/2$, and the companion form matches `paperI_main`. Table 2's five rows are covered by the frozen axiom file, which reported the standard footprint for all 15 of its declarations with exit code 0. 5. **Reused evidence is cited by direct path** and carries the required label: 20_EVIDENCE/H_LEAN_REUSE/results/ mirrors the external clean-room logs — the lake build log (Build completed successfully (8034 jobs), exit 0, 24m47.960s, zero errors), the frozen axiom run, and the auditor's own independent axiom query.

No Lean build was rerun. Per specification Section 5, this reuse does **not** carry over any manuscript, citation, bilingual, PDF or package verdict from v1.2 — every one of those was re-established fresh in this run.

10. Residual risks

Risk	Can this audit close it?
the printed citation URL does not resolve (expired third-party certificate)	No — the constant itself is now verified
erdosproblems.com returns HTTP 403 to this auditor, so the official problem page was never read directly	No — the open status is corroborated
novelty	No — this is a residual re-audit, not a new audit
PDFs not independently recompiled from the delivered TeX	No — producer strings and source-to-PDF are checked
the six project Lean modules outside the eight built targets	No — never compiled, no verdict
adequacy of the bespoke <code>PaperI.Split</code> structure as a model of every split graph	No — judged faithful by inspection; no formal proof

11. What this audit does not establish

- It does **not** establish that Theorem 1.1 is true; it establishes that a determined adversarial regression failed to falsify it.
- It is **not** human peer review.
- It does **not** prove global novelty.
- It does **not** re-derive the Lean development; Gate H is reused under byte identity.
- A `PASS_WITH_RESIDUALS` does **not** close any release gate that explicitly requires one of the residuals in Section 10.

12. Signature

Item	Value
Auditor	Claude Opus 5 (claude-opus-5), Anthropic
Configuration	primary auditor; the v1.2 run additionally used an adversarial challenger of the same model family
Audit date	2026-08-21
Specification	RESIDUAL_AUDIT_REQUEST_SPEC.md, SHA-256 e8d92cf5...5435
Target	preprint_draft_v1.3, EN Markdown f3094b67...5bea
Verdict	PASS_WITH_RESIDUALS — all 9 controls pass; general regression passes; 0 blocker, 0 major, 2 open minor, 1 note

The v1.2 report is **not** rewritten, relabelled or superseded by this document. It stands with its own **FAIL** verdict against its own hashes, and no evidence from the two runs is merged.